AIGenUT: Intelligent Unit Test Automation

Automating coverage, enforcing team style, and validating with absolute confidence.

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Testing shouldn’t be a bottleneck

Writing and maintaining manual unit tests results in low coverage, inconsistent styles, and tedious log-reading.

`AIGenUT` acts as your intelligent co-pilot, designed specifically to build high-quality, style-consistent test suites without the friction.

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The AIGenUT Architecture

A targeted, intelligent, and secure three-step workflow.

Local Visual Detection

Identify missing coverage.
Scan files locally.
No upload needed.

AI Generation via Skills

Synthesize unit tests.
Leverage skill modules.
Privacy focused generation.

Remote Re-Validation

Confirm functionality.
Execute secure checks.
Ensure robust results.

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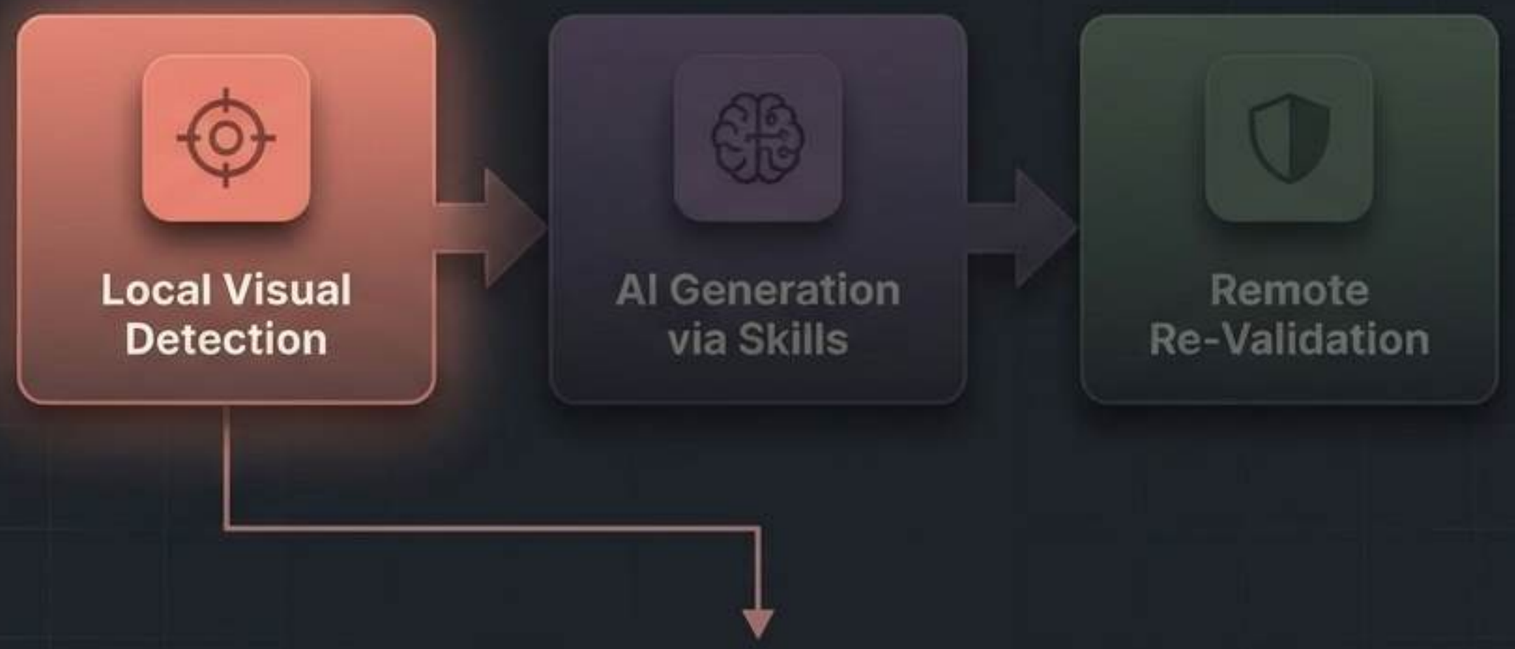

Step 1: Spot the weak areas without digging through logs.

A purely visual approach to codebase health. Identify exactly what `classes` and `methods` need attention before invoking the AI.

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Local Visual Detection → AI Generation via Skills → Remote Re-Validation

The Macro View: Instant Visibility

See the health of your codebase at a glance. We eliminate the need to parse complex test logs by surfacing file, method, and line coverage in a clean visual dashboard.

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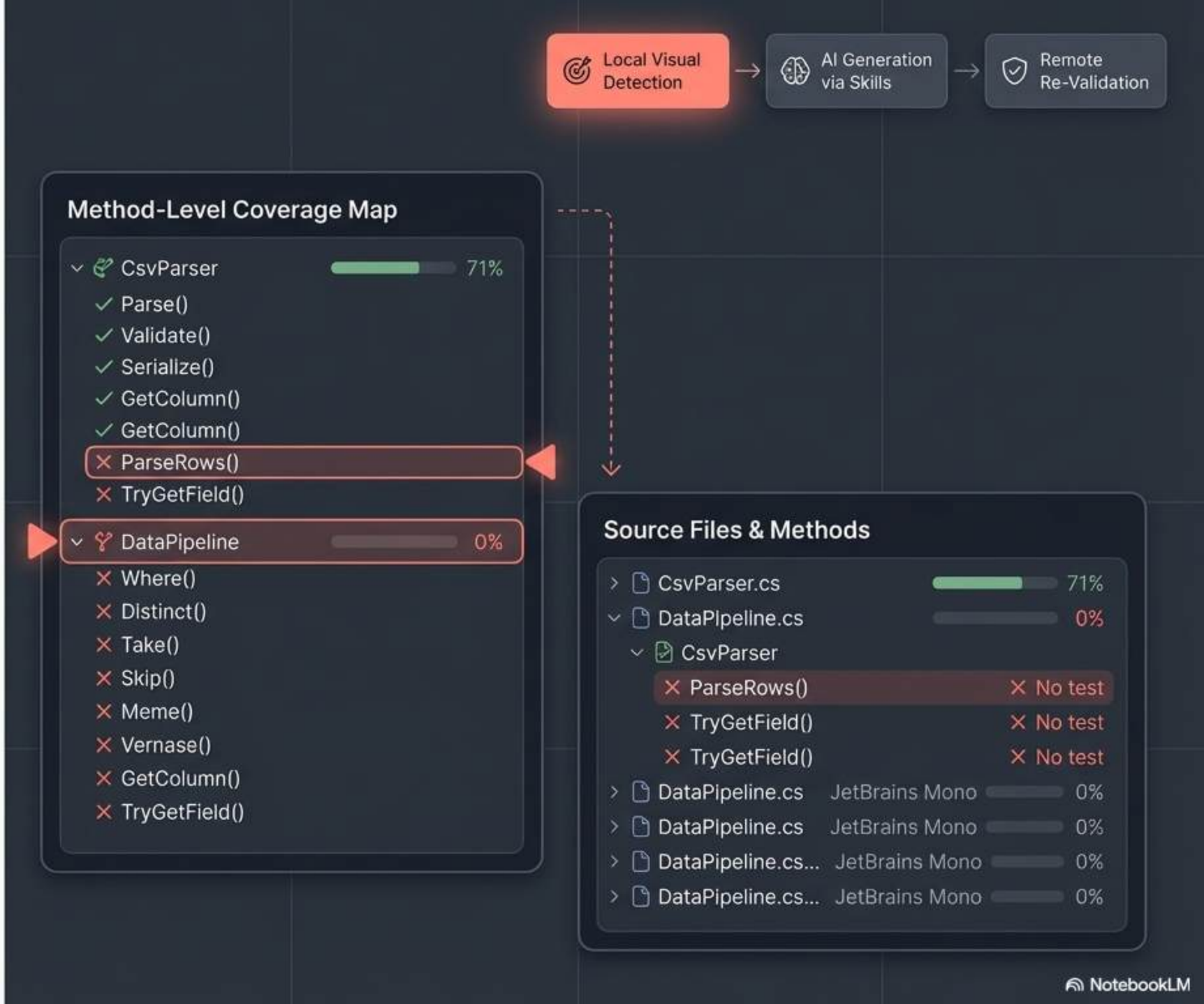
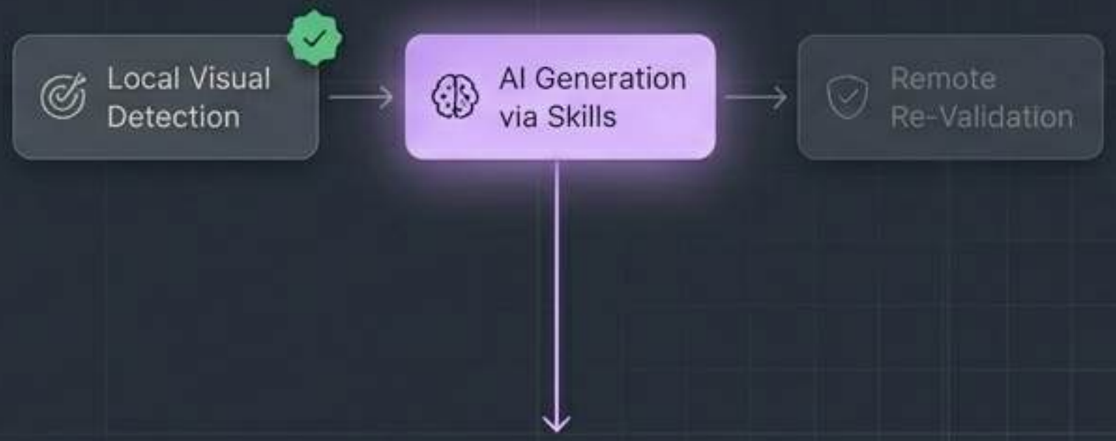
The Micro View: Pinpoint the Gaps

Green means tested. Red means missing. AIGenUT maps your codebase at the method level, telling the AI exactly where to focus its generation efforts.





Step 2: AI generation that understands your team's DNA.

Moving beyond generic AI to produce targeted, style-matched code.

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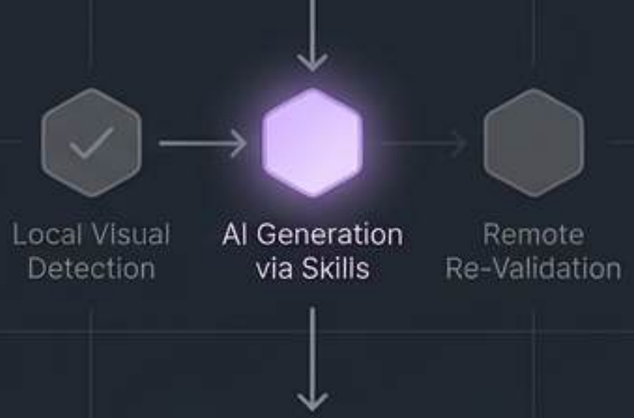
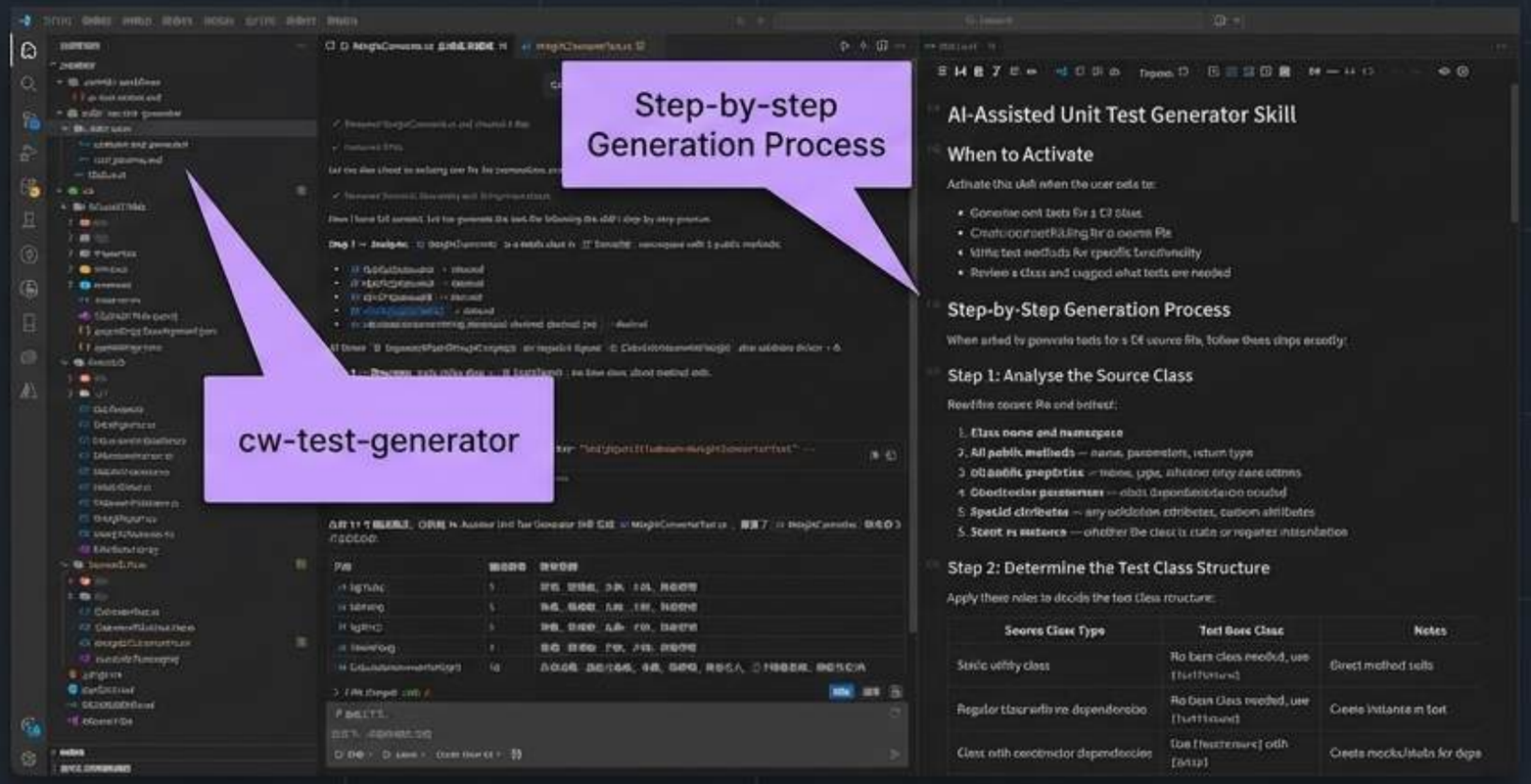


The Engine: What is a Skill?

A Skill is a curated set of rules and patterns. It teaches the AI to write tests in your specific style (e.g., CargoWise) rather than generic boilerplates—enforcing strict naming conventions and structural integrity.



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The Skill Advantage

Results in a 30% to 50% improvement in both coverage and style matching.

AIGenUT with Skills

```
Yes, 2026-03-08 | Features (Doc and one other)
using NUnit.Framework;

Yes, 2026-03-08 | Features (Doc and one other)
namespace Beerlib.Tests;

/// <summary>
/// Unit tests for WeightConverter - covers all public conversion methods
/// including happy path, edge cases, error cases, and round-trip conversions.
/// </summary>
Yes, 2026-03-08 | Features (Doc and one other)
[TestFixture]
0 references
public class WeightConverterTest
{
    // ------------------------------------------------------------
    // Kg to lbs
    // ------------------------------------------------------------

    [Test]
    0 references
    public void TestKgToLbs_WhenZero_ThenReturnsZero()
    {
        var result = WeightConverter.KgToLbs(0m);
        Assert.That(result, Is.EqualTo(0m));
    }

    [Test]
    0 references
    public void TestKgToLbs_WhenOneKg_ThenReturnsCorrectLbs()
    {
        var result = WeightConverter.KgToLbs(1m);
        Assert.That(result, Is.EqualTo(2.2046m));
```

- Targets identified gaps, matches CargoWise style perfectly.

Up to 50% Improvement

Standard AI

```
using Beerlib;
using NUnit.Framework;

namespace Beerlib.Tests;

[TestFixture]
0 references
public class WeightConverterTests
{
    #region Kg to lbs

    [Test]
    0 references
    public void KgToLbs_Zero_ReturnsZero()
    {
        Assert.That(WeightConverter.KgToLbs(0m), Is.EqualTo(0m));
    }

    [Test]
    0 references
    public void KgToLbs_PositiveValue_ReturnsCorrectResult()
    {
        // 1 kg = 2.20462 lbs (rounded to 4 decimals)
        Assert.That(WeightConverter.KgToLbs(5m), Is.EqualTo(11.0230m));
    }

    [Test]
    0 references
    public void KgToLbs_LargeValue_ReturnsCorrectResult()
    {
        // 100 kg = 220.462 lbs - rounded to 220.4620
        Assert.That(WeightConverter.KgToLbs(100m), Is.EqualTo(220.4620m));
    }
```

- Generic tests, low coverage, ignores team conventions.

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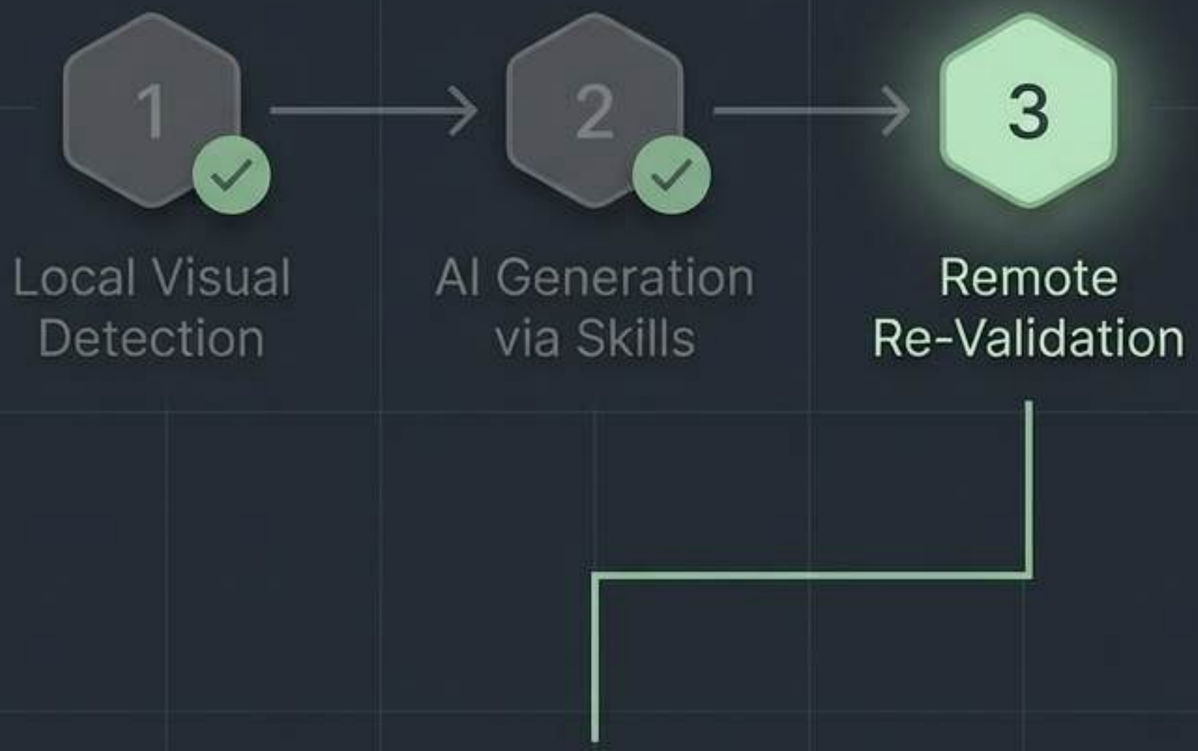

Step 3: The Safety Net. Because we don't trust blindly.

Local generation is fast, but `remote validation` guarantees absolute confidence before you merge.

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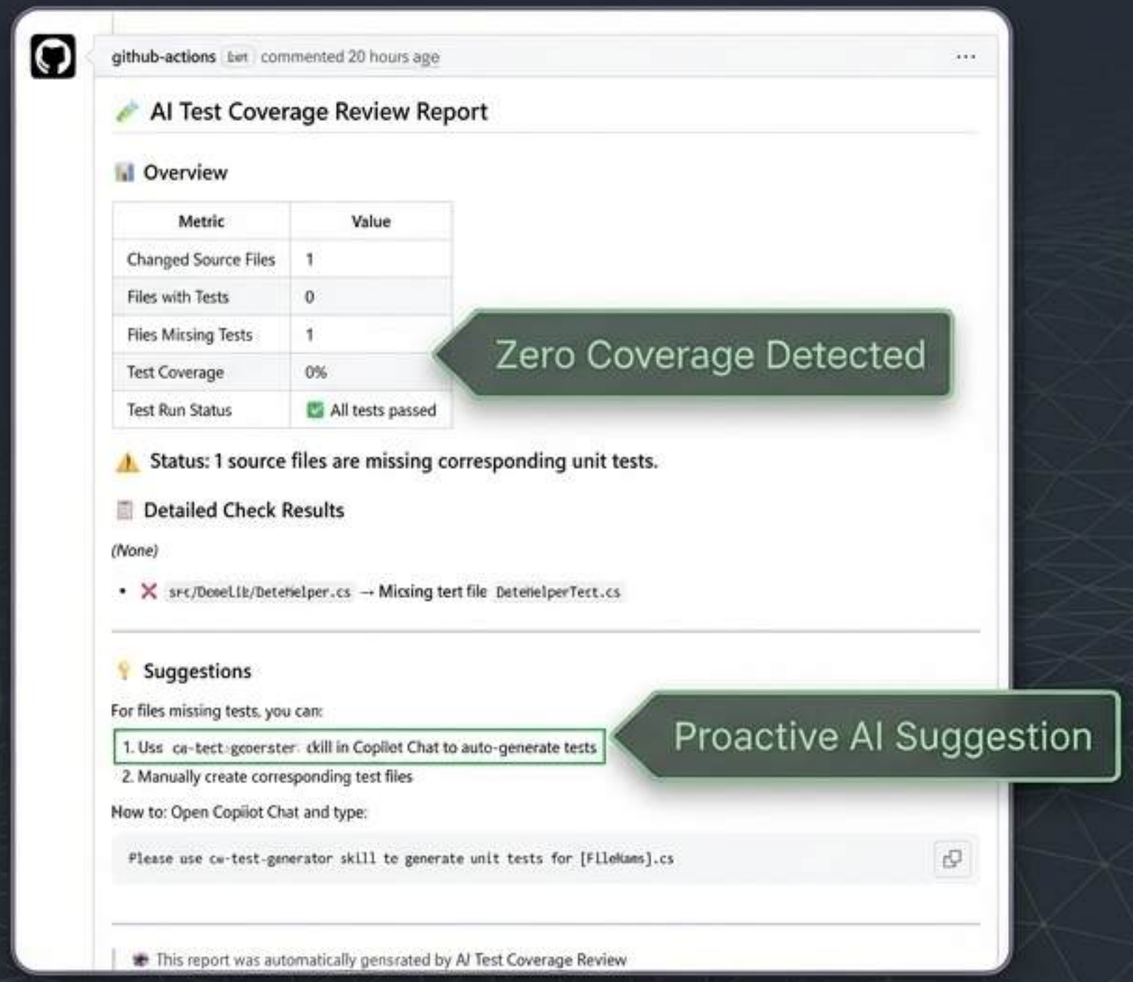
Remote Re-Validation via GitHub Actions

Push your code to GitHub, and our pipeline recalculates coverage in a clean, remote environment. It validates local numbers, catches missed files, and proactively suggests AI fixes directly in the PR.



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The Precision Pipeline Complete

Local Visual Detection

Identify the gaps locally.

AI Generation via Skills

Generate code intelligently with Skills.

Remote Re-Validation

Validate remotely with total confidence.

Three steps to flawless code coverage.

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Transform Your Testing Workflow

Ready to experience intelligent test automation? Explore the codebase, review the Skills integration, and run the pipeline yourself.

```
>_ https://github.com/Eden-dot-Ye/AIGenUT
```

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